MtRBD: Advancing iRBD Analysis With Multi-Task Learning for Joint Sleep Staging and RSWA Detection

Xiaoyu Chen, *Graduate Student Member, IEEE*, Yiyuan Zhang, Zhenning Tang, Huijuan Wu, Chen Chen, and Wei Chen, *Senior Member, IEEE*

***Abstract*—Rapid eye movement (REM) sleep without atonia (RSWA) is a critical diagnostic criterion for REM sleep behavior disorder (RBD). Current clinical practices rely on time-consuming manual annotation, increasing workload, and introducing variability. Existing automated methods, including CNN, LSTM, Transformer, and their combinations, fail to fully exploit the inherent physiological relationship between sleep staging and RSWA detection, while also facing challenges such as severe data imbalance and the complex fusion of multichannel signals. To address these limitations, we propose a multi-task learning framework that jointly optimizes both tasks. Our Multi-scale Information Attention Bottleneck Network (MIABNet) backbone improves multichannel fusion. Building upon MIABNet, the Multi-task RBD Network (MtRBD) implements dynamic feature enhancement, facilitating cross-task information flow while preserving physiological relationships. On the clinical CZ-RBD dataset collected from 30 patients over 59 nights at Shanghai Changzheng Hospital, our framework achieved the accuracy of 83.0% for sleep staging and 93.6% for RSWA detection, with an accuracy of 77.6% for critical RSWA event identification, outperforming single-task methods in cross-subject validation. Through modality selection and Grad-CAM visualization, we enhance clinical interpretability, providing reliable support for early detection of RBD and related neurodegenerative diseases.**

***Index Terms*—Rapid eye movement (REM) sleep behavior disorder, multi-task learning, sleep staging, REM sleep without atonia, information bottleneck.**

Received 23 May 2025; revised 10 August 2025; accepted 29 August 2025. Date of current version 8 December 2025. This work was supported by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 82302343. *(Corresponding authors: Huijuan Wu; Chen Chen; Wei Chen.)*

Xiaoyu Chen, Yiyuan Zhang, and Zhenning Tang are with the Department of Biomedical Engineering, College of Biomedical Engineering, Fudan University, Shanghai 200433, China (e-mail: xiaoyuchen23@m.fudan.edu.cn; yamsunshine1992@outlook.com; 22110720091@m.fudan.edu.cn).

Huijuan Wu is with the Department of Neurology, Second Affiliated Hospital of Naval Medical University, Shanghai 200003, China (e-mail: huijuan.w@163.com).

Chen Chen is with the Center for Medical Research and Innovation, Shanghai Pudong Hospital; Human Phenome Institute, Fudan University, Shanghai 201203, China (e-mail: chenchen_fd@fudan.edu.cn).

Wei Chen is with the School of Biomedical Engineering, University of Sydney, Camperdown, NSW 2050, Australia (e-mail: wei.chenbme@sydney.edu.au).

Digital Object Identifier 10.1109/JBHI.2025.3606944

I. Introduction

Rapid Eye Movement (REM) Sleep Behavior Disorder (RBD) is a sleep disorder characterized by abnormal dream-enacting behaviors and limb movements during REM sleep due to the failure of normal muscle atonia [1], [2]. During normal REM sleep, muscle tone is significantly reduced through inhibitory neurons in the brainstem, whereas RBD patients experience dysfunction in this inhibitory mechanism. REM Sleep Without Atonia (RSWA), defined as persistent or intermittent muscle activity during REM sleep as observed in electromyographic recordings, constitutes the core pathophysiological feature and primary diagnostic biomarker of RBD [2]. The isolated form of RBD (iRBD) serves as an important early marker of neurodegenerative diseases, often preceding formal diagnosis by 10–15 years [3]. Furthermore, longitudinal studies have shown that over 80% of iRBD patients eventually develop α-synucleinopathies, including Parkinson's disease, Lewy body dementia, or multiple system atrophy, highlighting the critical importance of timely detection of RBD [3], [4], [5], [6].

The current diagnostic protocol for RBD relies on overnight video-polysomnography (PSG) recordings, which requires specialists to visually identify REM sleep periods and RSWA through analysis of multiple biosignal categories, including electroencephalography (EEG), electrooculogram (EOG), and electromyography (EMG) [7], [8], [9]. Additionally, this approach is time-consuming and susceptible to subjective interpretation due to the lack of standardized criteria for muscle activity assessment. According to the AASM manual, human sleep is classified into five stages: Wake, N1, N2, N3, and REM [10]. RSWA detection categorizes each epoch as normal REM, abnormal REM (with RSWA), or non-REM. Accurate sleep staging is fundamental to RBD diagnosis as it identifies periods where RSWA may occur. Effective RSWA detection provides valuable disease assessment and monitoring indicators, driving the need for automated detection technologies.

With the advancement of artificial intelligence technologies such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Transformer architectures, numerous attempts have been made to automate RBD-related diagnostic analysis. Katarina et al. [7] pioneered the application of Spectral Vision Transformer (SViT) to PSG signals, demonstrating that EEG and EOG signals hold high diagnostic relevance for RBD, but their approach merely outputs a binary diagnostic decision without clinical interpretability. Navin et al. [8] employed random forest classifiers requiring tedious feature engineering (156 features from EEG, EOG, and EMG). Chen et al. [11] achieved promising results by fusing spatial and temporal graphs, validating them on datasets from healthy subjects and RBD patients. In addition, some methods focus on utilizing multi-channel and multi-modal signals for sleep disorder validation. For instance, Pan et al. [12] proposed MultiConsSleepNet, a self-supervised contrastive learning framework for sleep staging, which performs excellently with limited labeled clinical data. Zhou et al. [13] proposed the pseudo-siamese network PSEENet, which uses EEG and EOG for sleep staging, remaining stable even in the presence of heterogeneous signals or missing modalities, and outperforming existing methods on clinical sleep disorder datasets. However, like other existing methods, these approaches treat sleep staging and disease diagnosis as separate sequential tasks, where diagnostic accuracy heavily depends on the quality of sleep staging results.

Despite recent advances in sleep analysis techniques, RBD diagnosis remains challenging. Existing automated methods, such as traditional machine learning and deep learning models (e.g., CNNs and Transformers), struggle to effectively capture the physiological relationship between sleep staging and RSWA detection. These methods face issues like disjointed sleep staging and abnormal event detection [1], data imbalance (with REM periods making up only 15–25% of sleep time) [10], and complex multimodal signal integration [12], [13]. Traditional algorithms also perform poorly on RBD patients, particularly in distinguishing between REM and N1 periods due to overlapping EEG features. Additionally, research often separates sleep staging and RSWA detection, creating error propagation that affects RSWA detection accuracy. To date, no unified framework has effectively addressed these interrelated challenges.

To address these limitations, we propose an innovative multi-task learning framework that integrates sleep staging with RSWA detection. The main contributions of this paper are summarized as follows:

1) *Multi-scale Information Attention Bottleneck Network (MIABNet):* Developing a specialized backbone that effectively fuses multichannel PSG signals across temporal scales and modalities, demonstrating superior performance on both public sleep datasets and a real-world clinical CZ-RBD dataset.
2) *Unified Multi-task Framework (MtRBD):* The first framework to jointly optimize sleep staging and RSWA detection, leveraging their inherent physiological relationship to reduce error propagation, achieving 83.0% accuracy for sleep staging and 93.6% for RSWA detection.
3) *Dynamic Feature Enhancement Module (DFEM):* Introducing a novel module that adaptively adjusts feature importance based on sleep stage probabilities, effectively addressing severe data imbalance in RSWA detection and improving sensitivity to rare events, achieving 77.6% accuracy for critical RSWA events.
4) *Clinical Interpretability:* Enhancing practical utility through systematic modality contribution analysis and Grad-CAM visualization, providing transparent and clinically relevant insights for medical practitioners.

The remainder of this article is organized as follows: Section II describes the MIABNet and MtRBD; Section III presents the experimental setup and results; Section IV discusses our findings and limitations; and Section V concludes this article.

II. Methods

In this study, we first proposed a robust backbone network (MIABNet) based on a multi-scale information attention bottleneck structure. Building upon MIABNet, we further developed a multi-task learning network (MtRBD) for iRBD. The process of our network architecture is shown in Fig. 1.

A. Problem Formulation and Notations

Given a multimodal sleep recording dataset $\mathcal{D} = (\mathbf{X}_s, \mathbf{y}_s^s, \mathbf{y}_s^r), s = 1, 2, \ldots, N$, where N denotes the total number of 30-second samples, each sample $\mathbf{X}_s \in \mathbb{R}^{C\times L}$ contains C channels of physiological signals with L time points. Each sample is labeled with two different labels: $\mathbf{y}_s^s$ represents the sleep stage classification (Wake, N1, N2, N3, or REM), while $\mathbf{y}_s^r$ denotes the RSWA detection outcome (Non-REM, Normal REM, or Abnormal REM with RSWA).

Our objective is to develop a multi-task framework that simultaneously performs sleep staging (Task 1) and RSWA detection (Task 2).

B. Multi-Scale Information Attention Bottleneck Network (MIABNet)

1) Multi-Scale CNN Feature Extraction Module: To capture features across different time scales, we design a Multi-scale Convolutional Neural Network (MSCNN) architecture with three parallel convolutional branches. As presented in (1), each branch consists of three CNN units sequentially connected, along with max pooling layers to reduce dimension. Each unit is integrated with batch normalization and ReLU. The multi-scale design captures diverse temporal patterns crucial for distinguishing different sleep features [14].

$$F_{\text{branch}_i} = \text{Conv}_3(\text{MaxPool}(\text{Conv}_2(\text{MaxPool}(\text{Conv}_1(\mathbf{X}_s))))). \tag{1}$$

where $i \in \{1, 2, 3\}$ indexes the three parallel branches and $k_i \in \{3, 5, 7\}$ is the temporal kernel size assigned to branch i. Conv_j ($j \in \{1, 2, 3\}$) denotes the j-th convolutional unit in branch i, each followed by temporal max pooling.

2) Squeeze-and-Excitation Enhancement: Even though the multi-branch architecture extracts rich multi-scale features, the output inevitably contains some redundant feature information. To optimize the weight distribution of these features, we employ a Squeeze-and-Excitation (SE) block for each branch. Given an input feature $F_{\text{branch}_i} \in \mathbb{R}^{TD\times F\times C}$ from branch i, where TD represents the temporal dimension, F the frequency dimension, and C the channel dimension [9], [10]. The SE block operates as follows:

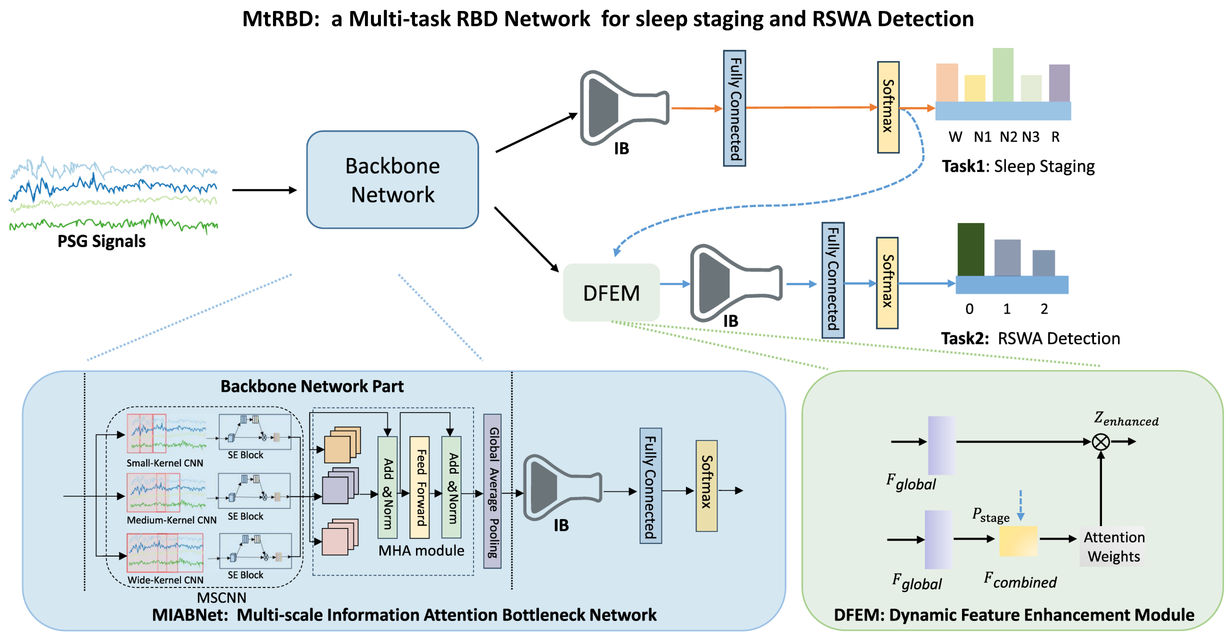

Fig. 1. Architecture of MtRBD: a multi-task network integrating sleep staging (orange arrows) and RSWA detection (blue arrows) with shared backbone (black modules). Sleep stage information flows to the RSWA pathway (dashed blue arrows). For Task 2 (RSWA Detection), the labels 0, 1, and 2 represent Non-REM, Normal REM, and Abnormal REM with RSWA.

First, global average pooling is applied to capture channel-wise global information $\mathbf{Z}_i$:

$$Z_i = \text{AvgPooling}(F_{\text{branch}_i}) \in \mathbb{R}^{1\times1\times C}, \tag{2}$$

Then, two fully connected layers with a ReLU activation in between are used to parameterize the feature selection mechanism :

$$S_i = \sigma(F_2(\text{ReLU}(F_1(Z_i)))) \in \mathbb{R}^{1\times1\times C}, \tag{3}$$

where F_1 reduces the dimension and F_2 restores it to C; $\sigma(\cdot)$ denotes the sigmoid activation function.

Finally, the channel-wise weights are applied to the original features:

$$F_{\text{SE}i} = S_i \times F\text{branch}_i, \tag{4}$$

After enhancing each branch's features, we integrate information across scales by concatenating the SE-enhanced features along the channel dimension:

$$F_{\text{concat}} = [F_{\text{SE}_1}, F_{\text{SE}_2}, F_{\text{SE}_3}], \tag{5}$$

Then apply a convolutional layer for dimensionality reduction, followed by batch normalization and ReLU activation:

$$F_{\text{reduced}} = \text{ReLU}(\text{BN}(\text{Conv1} \times 1(F_{\text{concat}}))). \tag{6}$$

This operation reinforces important features while suppressing less useful ones, effectively performing feature recalibration. By applying SE blocks to each branch separately, we enhance the distinctive temporal pattern learning capability of each scale before integration.

3) Multi-Headed Attention (MHA): To model temporal dependencies between sleep stages and capture the sequential nature of sleep patterns, we employ a Multi-Headed Attention (MHA) module. Given the input features $F_{\text{reduced}} \in \mathbb{R}^{N\times M}$, where N is the sequence length (number of time steps) and M is the feature dimension per time step, the MHA module generates Query ($Q \in \mathbb{R}^{N\times d_k}$), Key ($K \in \mathbb{R}^{N\times d_k}$), and Value ($V \in \mathbb{R}^{N\times d_v}$) matrices through linear transformations [15]:

$$Q = F_{\text{reduced}}W^Q, \quad K = F_{\text{reduced}}W^K, \quad V = F_{\text{reduced}}W^V. \tag{7}$$

where $W^Q \in \mathbb{R}^{M\times d_k}$, $W^K \in \mathbb{R}^{M\times d_k}$, and $W^V \in \mathbb{R}^{M\times d_v}$ are learnable parameter matrices, with d_k and d_v being the key/query dimension and value dimension respectively.

The self-attention mechanism computes weighted combinations of values based on the similarity between queries and keys:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{8}$$

where the scaling factor $\frac{1}{\sqrt{d_k}}$ ensures stable gradients during training by preventing the dot products from growing too large in magnitude.

The multi-head attention performs self-attention h times in parallel with different learned projections:

$$\text{head}_i = \text{Attention}(Q_i, K_i, V_i), \quad \text{for } 1 \le i \le h, \tag{9}$$

$$F_{\text{MHA}} = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)W^O. \tag{10}$$

where h is the number of attention heads, and $W^O \in \mathbb{R}^{h\cdot d_v\times M}$ is an output projection matrix that combines the concatenated attention outputs back to dimension M.

This integrated approach to cross-scale integration and temporal modeling creates a unified feature representation that preserves scale-specific characteristics while encoding complex temporal relationships.

4) Global Feature Extraction: After the multi-head attention module, we apply global average pooling to extract a sequence-level representation:

$$F_{\text{global}} = \text{AvgPooling}(F_{\text{MHA}}). \tag{11}$$

This operation compresses the temporal dimension and produces a fixed-length feature vector $F_{\text{global}} \in \mathbb{R}^D$, where D is the feature dimension. The global features serve as input to the information bottleneck module and provide a comprehensive representation of the physiological signals.

5) Information Bottleneck (IB): After global feature extraction, we apply a variational information bottleneck to enforce compressed representation [16], [17]. The encoder maps the input features to a probabilistic latent space characterized by mean μ and log-variance $\log \sigma^2$:

$$\mu = W_\mu F_{\text{global}} + b_\mu, \tag{12}$$

$$\log \sigma^2 = W_{\log \sigma^2} F_{\text{global}} + b_{\log \sigma^2}, \tag{13}$$

where $W_\mu \in \mathbb{R}^{d\times D}$, $W_{\log \sigma^2} \in \mathbb{R}^{d\times D}$, $b_\mu \in \mathbb{R}^d$, and $b_{\log \sigma^2} \in \mathbb{R}^d$ are learnable parameters, and d is the dimensionality of the bottleneck space.

The latent representation z is sampled using the reparameterization trick:

$$z = \mu + \epsilon \otimes \sigma, \quad \epsilon \sim \mathcal{N}(0, I), \tag{14}$$

where ϵ is random noise sampled from a standard normal distribution, and $\otimes$ represents element-wise multiplication. This introduces stochasticity while allowing gradient backpropagation.

The bottleneck constraint is enforced via a Kullback-Leibler (KL) divergence term added to the loss function:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2}\sum_{i=1}^{d}\left(1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2\right). \tag{15}$$

This regularization term encourages the model to learn a compact representation by penalizing deviations from a standard normal distribution, effectively compressing the feature space to retain only the most discriminative information.

6) MIABNet Loss Function: The overall loss function for the MIABNet model combines the task-specific cross-entropy loss for sleep staging with the KL divergence regularization term:

$$\mathcal{L}\text{MIABNet} = \mathcal{L}\text{task} + \lambda_0 \cdot \mathcal{L}_{\text{KL}}. \tag{16}$$

where $\mathcal{L}_{\text{task}}$ is the cross-entropy loss for sleep-related tasks, and λ_0 controls the strength of the information bottleneck constraint.

C. Multi-Task RBD Network (MtRBD)

The MtRBD framework extends MIABNet with a hierarchical multi-task architecture that incorporates probabilistic information flow by DFEM and task-specific information bottlenecks [18].

1) Task 1: Sleep Staging: The first task performs sleep staging using the global feature representation processed through a dedicated information bottleneck:

$$z_{\text{task1}}, L_{\text{KL1}} = \text{InformationBottleneck1}(F_{\text{global}}), \tag{17}$$

$$P_{\text{stage}} = \text{Softmax}(W_{\text{stage}} \cdot z_{\text{task1}} + b_{\text{stage}}), \tag{18}$$

where F_{global} represents the global features extracted from the backbone network, z_{task1} is the compressed representation from the first information bottleneck, and W_{stage} and b_{stage} are classifier parameters.

The loss function for Task 1 combines the cross-entropy loss for sleep staging with a sparsity-promoting reconstruction constraint:

$$\mathcal{L}\text{task1} = \mathcal{L}\text{stage} + \lambda_1 \cdot L_{\text{KL1}}. \tag{19}$$

where L_{KL1} is the KL divergence term from the first information bottleneck, and λ_1 controls the strength of the information bottleneck constraints

2) Dynamic Feature Enhancement Module (DFEM): A key innovation in our MtRBD model is the Dynamic Feature Enhancement Module that implements an attention mechanism to modulate global features based on sleep stage probabilities. This module generates feature-wise attention weights conditioned on sleep stage information:

$$F_{\text{combined}} = [F_{\text{global}}, P_{\text{stage}}], \tag{20}$$

$$\text{Attention}_{\text{weights}} = \sigma(W_2 \text{ReLU}(W_1 F_{\text{combined}})), \tag{21}$$

$$Z_{\text{enhanced}} = F_{\text{global}} \otimes \text{Attention}_{\text{weights}}. \tag{22}$$

where σ represents the sigmoid activation, $\otimes$ denotes element-wise multiplication, $W_1 \in \mathbb{R}^{H\times(D+P)}$ and $W_2 \in \mathbb{R}^{D\times H}$ are learnable weights (H is the hidden dimension, D is the feature dimension, and P is the dimension of the sleep stage probability information), and $F_{\text{combined}} \in \mathbb{R}^{D+P}$ combines the original global features with sleep stage probabilities.

This attention mechanism dynamically emphasizes the most task-relevant features based on predicted sleep stages, adaptively focusing on patterns most indicative of RSWA for the current sleep context. By applying attention to the original feature space rather than using the compressed representations from Task 1, we preserve the full richness of the signal for RSWA detection while still leveraging the predictive information from sleep staging.

3) Task 2: RSWA Detection: The enhanced features undergo a second task-specific information bottleneck to extract the most relevant features for RSWA detection:

$$z_{\text{task2}}, L_{\text{KL2}} = \text{InformationBottleneck2}(z_{\text{enhanced}}), \tag{23}$$

$$P_{\text{RBD}} = \text{Softmax}(W_{\text{RBD}} \cdot z_{\text{task2}} + b_{\text{RBD}}), \tag{24}$$

The loss function for RSWA detection also includes an information bottleneck constraint:

$$\mathcal{L}\text{task2} = \mathcal{L}\text{RBD} + \lambda_2 \cdot L_{\text{KL2}}. \tag{25}$$

where L_{KL2} is the KL divergence term from the second information bottleneck, and λ_2 controls the strength of the information bottleneck constraints

TABLE I
DETAILS OF THE SLEEP DATASET USED IN OUR EXPERIMENTS

Dataset Name	Numbers (Male, Female)	Age (mean ± std.dev.)	Sleep Staging Labels W	N1	N2	N3	REM	RSWA Detection Labels Non-REM	Normal REM	Abnormal REM with RSWA
SleepEDF-20	20 (10, 10)	28.65 ± 2.58 years	8285 (19.6%)	2804 (6.6%)	17799 (42.1%)	5703 (13.5%)	7717 (18.2%)	-	-	-
CZ-RBD	30 (19, 11)	66.23 ± 12.15 years	12919 (21.8%)	6665 (11.2%)	26900 (45.4%)	2995 (5.1%)	9810 (16.5%)	49479 (83.5%)	7238 (12.2%)	2572 (4.3%)

4) Total Loss Function: The overall loss function combines task-specific losses with their respective constraints:

$$\mathcal{L}\text{total} = \alpha \cdot \mathcal{L}_{\text{task1}} + \beta \cdot \mathcal{L}_{\text{task2}}. \tag{26}$$

where α and β are weighting coefficients that balance the importance of each task.

III. EXPERIMENTS AND RESULTS

A. Data and Preprocessing

The SleepEDF-20 dataset contains recordings from 20 healthy subjects, including 2 EEG channels, 1 EOG channel, 1 EMG channel, and 1 oronasal airflow channel, with a sampling rate of 100 Hz [19]. The dataset comprises 39 nights of recordings. To align with modern clinical practice, we converted the original sleep staging from the R&K standard to the five-stage AASM standard by merging N3 and N4 into a unified N3 stage and excluding invalid labels [10]. Following previous research [11], [13], we utilized EEG (Fpz-Cz and Pz-Oz) and horizontal EOG signals.

The CZ-RBD dataset was collected at Shanghai Changzheng Hospital with institutional ethics committee approval (ethics permit number: 2024SL081). All 30 iRBD patients provided informed consent after receiving comprehensive information about the study. The dataset includes PSG recordings from 59 nights(2 consecutive nights/participant), with one participant's first night (ID=30) missing due to technical failure. The acquired PSG data comprises 3 EEG channels (F4-A1, C4-A1, and O2-A1), 2 EOG channels (E1-M2 and E2-M2), 1 chin EMG channel, and bilateral tibialis anterior EMG channels. Each recording epoch was annotated by three board-certified sleep specialists for both sleep stages and RSWA detection. The expert panel held consensus meetings for disagreements, with difficult cases adjudicated by a senior neurosurgeon.

Detailed data statistics are presented in Table I. Signal preprocessing followed a standardized pipeline: notch filters eliminated 50 Hz power line interference, followed by channel-specific bandpass filtering: EEG signals at 0.3–50 Hz, EOG signals at 0.1–50 Hz, and EMG signals at 10–50 Hz. SleepEDF-20 retains original adoption rate; CZ-RBD all channels were downsampled from the original 200 Hz to 100 Hz and normalized within each subject to reduce inter-individual variability while preserving intra-recording dynamics.

B. Experimental Setup

The model evaluation was performed using leave-one-subject-out cross-validation (LOSO), in which data from one participant was used for testing, data from another participant was reserved for validation, and data from the remaining participants were used for training in each iteration. This validation set was used for early stopping and hyperparameter tuning during model development. For the SleepEDF-20 dataset, a 20-fold LOSO was applied to evaluate the sleep staging performance of the proposed MIABNet model in a single-task setting. For the CZ-RBD dataset, both single-task and multi-task evaluations were conducted using a 30-fold LOSO to comprehensively assess model performance.

During training, the model employed an MHA with $H = 16$ heads, task weight $\alpha = 0.5$ for sleep staging, and task weight $\beta = 0.5$ for RSWA detection. The information bottleneck dimension was set to $d = 64$, with constraint weights $\lambda_1 = 0.01$ and $\lambda_2 = 0.01$, respectively. These hyperparameters were determined through a comprehensive grid search to identify the optimal configuration. The Adam optimizer was applied with a learning rate of 0.001, a batch size of 128, and the training ran for 200 epochs. An early stopping criterion with a patience of 20 epochs was implemented to prevent overfitting. Evaluation metrics, following common practices in sleep task research, included accuracy, macro F1 score (MF1), Cohen's Kappa, and F1 scores for each class. The experiments were conducted on a workstation equipped with an NVIDIA RTX 4090 GPU, an Intel i9-14900K CPU, and 64GB of RAM. All models were implemented using PyTorch 2.4.1.

TABLE II
PERFORMANCE COMPARISON OF SLEEP STAGING METHODS ON SLEEPEDF-20 DATASET

Methods	ACC	MF1	κ	F1-W	F1-N1	F1-N2	F1-N3	F1-R
DeepSleepNet [20]	0.820	0.769	0.760	0.847	**0.466**	0.851	0.898	0.826
MRASleepNet [21]	0.845	0.789	0.786	0.907	0.461	0.887	0.882	0.807
AttnSleep [15]	0.844	0.781	0.790	0.897	0.426	**0.888**	**0.902**	0.790
PSEENet [13]	0.833	0.766	0.770	0.903	0.423	0.878	0.827	0.801
MIABNet	**0.853**	**0.792**	**0.798**	**0.922**	0.442	0.882	0.876	**0.837**

C. Results and Comparison

1) Performance on SleepEDF-20 Dataset: As shown in Table II, MIABNet achieved state-of-the-art performance on the SleepEDF-20 dataset, with an accuracy of 85.3% and a macro F1-score of 0.792. Compared with specialized sleep staging models such as DeepSleepNet, MRASleepNet, AttnSleep, and PSEENet [13], [15], [20], [21], MIABNet attained higher or comparable results across all key metrics. Notably, MIABNet achieved superior F1-scores in detecting Wake (0.922) and REM (0.837) stages, and obtained the highest Cohen's kappa value (0.798), reflecting reliable classification. The macro and

TABLE III
PERFORMANCE COMPARISON OF MTRBD, AND SINGLE-TASK MODELS ON THE CZ-RBD DATASET

Model	Sleep Staging								RSWA Detection					
	ACC	MF1	κ	F1-W	F1-N1	F1-N2	F1-N3	F1-REM	ACC	MF1	κ	F1-0	F1-1	F1-2
Sleepyco [22]	0.791	0.707	0.728	0.868	0.316	0.845	0.774	0.731	–	–	–	–	–	–
SleepHGCN [23]	0.784	0.730	0.711	0.823	0.458	0.862	0.767	0.742	–	–	–	–	–	–
CNN+Attention [18]	–	–	–	–	–	–	–	–	0.913	0.728	0.693	0.961	0.732	0.490
CNN+LSTM [18]	–	–	–	–	–	–	–	–	0.911	0.749	0.691	0.958	0.730	0.560
Deepsleepnet+ [20]	0.638	0.538	0.447	0.465	0.465	0.769	0.466	0.523	0.866	0.597	0.555	**0.975**	0.447	0.322
Attensleep+ [15]	0.813	0.746	0.731	0.872	0.501	0.863	0.712	0.801	0.915	0.737	0.697	0.961	0.735	0.514
PSEENet+ [13]	0.811	0.732	0.729	0.865	0.471	0.880	0.674	0.794	0.909	0.720	0.700	0.963	0.736	0.462
ASTGSleep+ [11]	0.823	0.758	0.747	0.880	0.513	0.885	0.708	0.803	0.920	0.774	0.732	0.963	0.761	0.598
MIABNet (task1)	0.818	0.745	0.739	0.884	0.481	0.878	0.621	0.809	–	–	–	–	–	–
MIABNet (task2)	–	–	–	–	–	–	–	–	0.917	0.760	0.704	0.965	0.698	0.615
MtRBD	**0.830**	**0.771**	**0.756**	**0.885**	**0.515**	**0.885**	**0.754**	**0.816**	**0.936**	**0.824**	**0.782**	0.969	**0.797**	**0.705**

Note: For Task 2 (RSWA Detection), the labels 0, 1, and 2 represent Non-REM, Normal REM, and Abnormal REM with RSWA. The same is applied to the other tables.

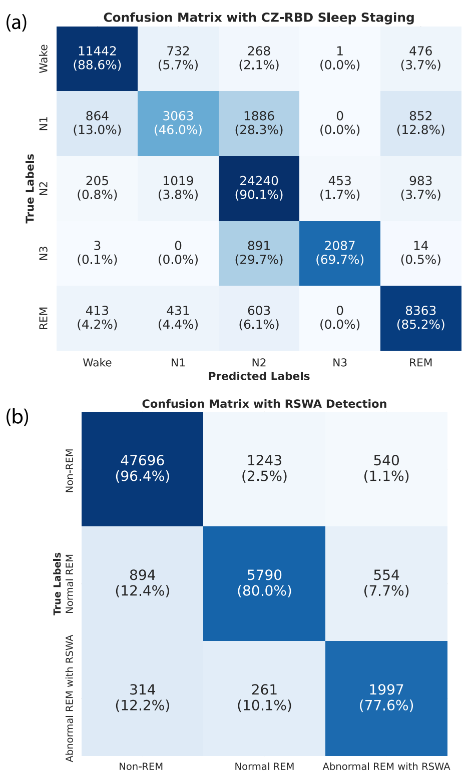

Fig. 2. Confusion matrix of MtRBD for CZ-RBD. The values in parentheses indicate the proportion. (a) Represents sleep staging, and (b) represents RSWA detection.

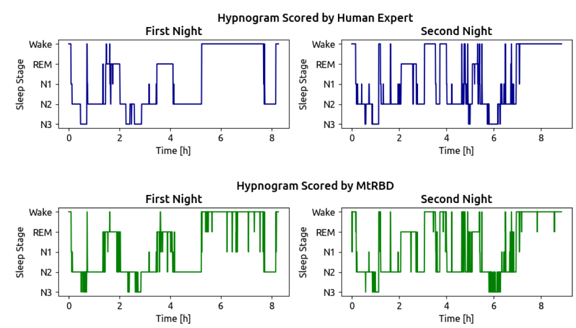

Fig. 3. Hypnogram scored by a human expert (top) and the hypnogram scored by MtRBD (bottom) for Subject ID:14.

per-class F1-scores further indicate the model's advantage in identifying transitions between different sleep states.

These results demonstrate that MIABNet not only advances the overall accuracy of automatic sleep staging but also delivers more consistent and reliable performance across various sleep stage transitions, particularly at critical boundaries between wakefulness and sleep states, further affirming its applicability for sleep analysis.

2) Performance on CZ-RBD Clinical Dataset: We evaluated both single-task and multi-task approaches on the more challenging clinical CZ-RBD dataset, as shown in Fig. 2. Table III presents the comprehensive performance comparison across both sleep staging and RSWA detection tasks.

For sleep staging, the MtRBD framework achieved an accuracy of 83.0% and a Cohen's kappa of 0.756, with particularly high accuracy in classifying N2 (F1-N2: 0.885) and REM (F1-REM: 0.816) stages. This represents a significant improvement over its single-task variant (MIABNet: 81.8% accuracy, 0.745 MF1, 0.739 kappa), and substantially outperforms multi-task learning sleep staging networks developed using classic state-of-the-art sleep networks as a backbone, such as DeepSleepNet+ (accuracy 63.8%), AttenSleep+ (accuracy 81.3%), PSEENet+ (accuracy 81.1%), and ASTGSleep+ (accuracy 82.3%). Fig. 3 demonstrates that MtRBD consistently tracks sleep patterns across continuous and fragmented sleep nights for a representative subject (ID: 14), indicating stable performance under varying sleep conditions.

For RSWA detection, MtRBD achieved a high overall accuracy of 93.6% and a macro F1-score of 0.824, with a 0.705 F1-score in detecting Abnormal REM with RSWA, the critical diagnostic biomarker for RBD. This performance significantly outperformed its single-task counterpart (MIABNet: 91.7% accuracy, 0.760 MF1) and baseline models, including CNN+Attention (91.3% accuracy, 0.728 MF1) and CNN+LSTM (91.1% accuracy, 0.749 MF1). The confusion matrices in Fig. 2(b) illustrate the model's exceptional discrimination between normal and abnormal REM episodes with minimal

TABLE IV
VALIDATION OF MODALITIES IN THE MTRBD MODEL

Model	Sleep Staging		
	ACC	MF1	κ
Only MSCNN	0.826	0.758	0.762
MIABNet w/o SE	0.850	0.792	0.795
MIABNet w/o MHA	0.841	0.776	0.783
MIABNet w/o IB	0.830	0.759	0.765
MIABNet	**0.853**	**0.792**	**0.798**

diagnostic confusion. MtRBD achieved F1 scores of 0.797 for normal REM and 0.705 for abnormal REM, considerably higher than AttenSleep+ (0.735, 0.514), ASTGSleep+ (0.761, 0.598), and other baseline networks.

These results highlight the synergistic effect of the multi-task framework, where joint learning of sleep staging and RSWA detection mutually reinforces performance on both tasks. Additionally, the DFEM architecture helps the model focus on key features in both tasks.

D. Ablation Study

To assess the contribution of each component, we performed ablations on MIABNet and MtRBD. In MIABNet, removing SE, MHA, or IB consistently degraded performance, with IB having the largest impact by constraining the information bottleneck to suppress nuisance variability and enhance discrimination. In MtRBD, the full model outperformed all partial variants, with the largest gains observed in RSWA detection. DFEM leverages sleep-stage context to refine RSWA decisions, and the information bottleneck yields additional improvements by preserving task-relevant features. A module-wise breakdown and sensitivity analyses are provided in the Supplementary Materials (S.2 Ablation Study).

Further details, including the algorithmic flowchart, Ablation Study, hyperparameter validations, and additional dataset validation, are provided in the Supplementary Materials (S.1–S.4 including S.2 Ablation Study) at https://github.com/GeorgeChenn1/MtRBD.

IV. DICUSSION

A. Modality Selection

The current literature recommends combining EEG, EOG, and EMG to capture comprehensive sleep-related physiology [4], [8]. Accordingly, we integrate all three modalities and evaluate them on the CZ-RBD dataset. As shown in Table IV, EEG signals alone can achieve a relatively high baseline performance for sleep staging, while adding EOG further enhances accuracy. The addition of EMG to EEG+EOG provides only marginal improvement in sleep staging but substantially improves RSWA detection, consistent with previous studies [4], [5], [8]. Clinically, single-channel signals alone are insufficient for reliable iRBD diagnosis, and multimodal, multichannel recordings remain essential for more robust and objective evaluations.

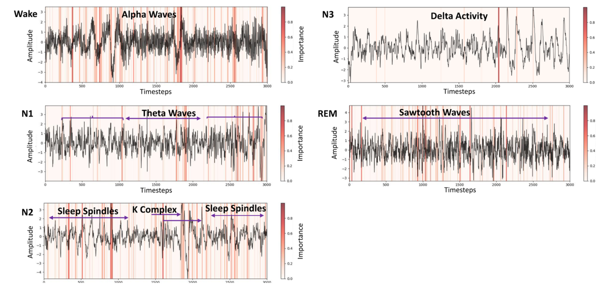

Fig. 4. The Grad-CAM output of MtRBD for sleep staging: model attention across different sleep stages.

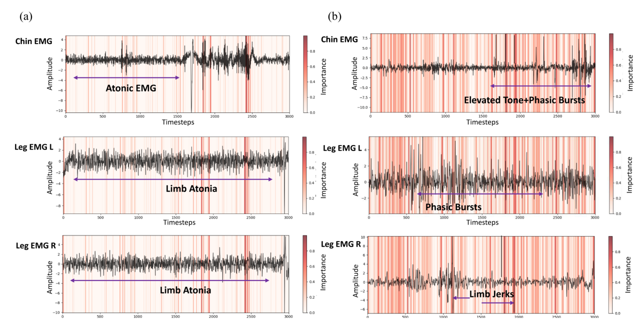

Fig. 5. The Grad-CAM output of MtRBD for RSWA detection. (a) Normal REM; (b) Abnormal REM with RSWA.

B. Model Interpretability

We employed Grad-CAM visualization to examine how the MtRBD model processes PSG signals (Figs. 4 and 5) [24], [25]. We selected the F4-A1 EEG channel for sleep staging visualization due to its higher attribution weights in our model and clear manifestation of stage-specific waveforms, while EMG channels were chosen for their critical role in capturing RBD's muscle activity abnormalities.

In EEG visualizations (Fig. 4), the model focuses on physiologically relevant features across sleep stages: alpha waves (8–12 Hz) during Wake, theta transitions (4–7 Hz) in N1, sleep spindles and K-complexes in N2, delta oscillations (0.5–4 Hz) in N3, and sawtooth waves with mixed-frequency activity during REM sleep. For EMG analysis (Fig. 5), we observed distinct attention patterns: in normal REM sleep (Fig. 5(a)), the model focuses on sustained low-amplitude segments indicating muscle atonia, while during abnormal REM episodes (Fig. 5(b)), attention shifts to elevated muscle tone, phasic bursts in chin EMG, and limb jerks.

These visualizations demonstrate the model's ability to identify task-specific physiological features, illustrating how our multi-task approach effectively integrates sleep staging and RSWA detection information to provide interpretable insights that could aid clinical diagnosis.

C. Limitations and Future Work

The proposed models in this study exhibit robust performance across both single-task and multi-task analyses related to RBD sleep disorders, but several limitations remain. Due to the low prevalence of RBD in elderly populations and the labor-intensive nature of PSG annotation, our dataset is relatively small and does not include iRBD patients with prodromal Parkinson's disease (PD) [26]. This limits the generalizability and clinical applicability of our findings. Another challenge is the class imbalance in RSWA detection, which arises from the randomness and individual variability of abnormal events and leads to a scarcity of abnormal REM samples. The DFEM partially mitigates this issue by incorporating sleep stage probabilities into feature representations, but further improvement is needed, particularly in detecting rare abnormal events and reducing missed detections under highly imbalanced conditions. Additionally, larger and more diverse datasets are necessary to validate the robustness and applicability of our model across different populations and clinical contexts. In future work, we plan to expand our dataset through multi-center collaboration, include cases with prodromal PD, and explore self-supervised pretraining on large-scale unlabeled PSG data to enhance model robustness, especially in low-data scenarios. Longitudinal studies tracking iRBD progression may also support the development of early prediction models for neurodegenerative disease conversion and improve timely clinical intervention.

V. Conclusion

In this study, we develop MtRBD, a multi-task framework to jointly perform sleep staging and RSWA detection from multichannel PSG signals. Within the model, the MIABNet backbone integrates multi-scale information bottleneck attention to effectively extract and fuse features across multiple modalities and temporal scales. By establishing a probabilistic information flow between sleep staging and RSWA detection, MtRBD dynamically guides feature attention and adaptively adjusts loss weights, effectively addressing the class imbalance problem while capturing the intrinsic association between the two tasks. This unified approach leads to significant performance improvements in both sleep staging and RSWA detection compared to single-task models. Our results demonstrate important clinical implications for the early identification of neurodegenerative disease risks. Furthermore, this framework holds promise for future development as a tool for quantifying RBD severity and advancing automated analysis of sleep disorders.

References

[1] G. Maggi, C. Vitale, F. Cerciello, and G. Santangelo, "Sleep and wakefulness disturbances in parkinson's disease: A meta-analysis on prevalence and clinical aspects of REM sleep behavior disorder, excessive daytime sleepiness and insomnia," *Sleep Med. Rev.*, vol. 68, 2023, Art. no. 101759.

[2] B. Huang et al., "Gut microbiome dysbiosis across early parkinson's disease, REM sleep behavior disorder and their first-degree relatives," *Nature Commun.*, vol. 14, no. 1, 2023, Art. no. 2501.

[3] S. Feuerstein et al., "Sleep structure discriminates patients with isolated REM sleep behavior disorder: A deep learning approach," in*Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2024, pp. 1–4.

[4] R. B. Postuma et al., "Risk and predictors of dementia and parkinsonism in idiopathic REM sleep behaviour disorder: A multicentre study," *Brain*, vol. 142, no. 3, pp. 744–759, 2019.

[5] G. Bruschi et al., "Tracking prodromal α-synucleinopathies: Novel fluid and tissue-based biomarker approaches for iRBD phenoconversion," *Sleep press*, 2025, Art. no. zsaf174.

[6] A. Iranzo et al., "Neurodegenerative disease status and post-mortem pathology in idiopathic rapid-eye-movement sleep behaviour disorder: An observational cohort study," *Lancet Neurol.*, vol. 12, no. 5, pp. 443–453, 2013.

[7] K. M. Gunter, A. Brink-Kjær, E. Mignot, H. B. D. Sørensen, E. During, and P. Jennum, "SViT: A spectral vision transformer for the detection of REM sleep behavior disorder," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 9, pp. 4285–4292, Sep. 2023.

[8] N. Cooray et al., "Detection of REM sleep behaviour disorder by automated polysomnography analysis," *Clin. Neuriophysiol.*, vol. 130, no. 4, pp. 505–514, 2019.

[9] M. Figorilli et al., "Neurophysiological aspects of REM sleep behavior disorder (RBD): A narrative review," *Brain Sci.*, vol. 11, no. 12, 2021, Art. no. 1588.

[10] R. B. Berry et al., "Rules for scoring respiratory events in sleep: Update of the 2007 AASM manual for the scoring of sleep and associated events: Deliberations of the sleep apnea definitions task force of the american academy of sleep medicine," *J. Clin. Sleep Med.*, vol. 8, no. 5, pp. 597–619, 2012.

[11] X. Chen et al., "ASTGSleep: Attention based spatial-temporal graph network for sleep staging," *IEEE Trans. Instrum. Meas.*, vol. 74, 2025, Art. no. 4004214.

[12] J. Pan et al., "A multimodal consistency-based self-supervised contrastive learning framework for automated sleep staging in patients with disorders of consciousness," *IEEE J. Biomed. Health Informat.*, vol. 29, no. 2, pp. 1320–1332, Feb. 2025.

[13] W. Zhou et al., "PSEENet: A pseudo-siamese neural network incorporating electroencephalography and electrooculography characteristics for heterogeneous sleep staging," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 9, pp. 5189–5200, Sep. 2024.

[14] H. Zhu et al., "MaskSleepNet: A cross-modality adaptation neural network for heterogeneous signals processing in sleep staging," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 5, pp. 2353–2364, May 2023.

[15] E. Eldele et al., "An attention-based deep learning approach for sleep stage classification with single-channel EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 809–818, 2021.

[16] N. Tishby, F. C. Pereira, and W. Bialek, "The information bottleneck method," 2000, *arXiv:physics/0004057*.

[17] S. Hu et al., "IPCT-Net: Parallel information bottleneck modality fusion network for obstructive sleep apnea diagnosis," *Neural Netw.*, vol. 181, 2025, Art. no. 106836.

[18] L. Cheng,S. Luo,X. Yu,H. Ghayvat,H. Zhang, and Y. Zhang, "EEG-CLNe: Collaborative learning for simultaneous measurement of sleep stages and OSA events based on single EEG signal," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 2503910.

[19] B. Kemp, A. H. Zwinderman, B. Tuk, H. A. Kamphuisen, and J. J. Oberye, "Analysis of a sleep-dependent neuronal feedback loop: The slow-wave microcontinuity of the EEG," *IEEE Trans. Biomed. Eng.*, vol. 47, no. 9, pp. 1185–1194, Sep. 2000.

[20] A. Supratak, H. Dong, C. Wu, and Y. Guo., "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 11, pp. 1998–2008, Nov. 2017.

[21] R. Yu, Z. Zhou, S. Wu, X. Gao, and G. Bin, "MRASleepNet: A multi-resolution attention network for sleep stage classification using single-channel EEG," *J. Neural Eng.*, vol. 19, no. 6, 2022, Art. no. 066025.

[22] S. Lee et al., "Sleepyco: Automatic sleep scoring with feature pyramid and contrastive learning," *Expert Syst. Appl.*, vol. 240, 2024, Art. no. 122551.

[23] Z. Jia et al., "Exploiting interactivity and heterogeneity for sleep stage classification via heterogeneous graph neural network," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2023, pp. 1–5.

[24] M. Dutt et al., "SleepXAI: An explainable deep learning approach for multi-class sleep stage identification," *Appl. Intell.*, vol. 53, no. 13, pp. 16830–16843, 2023.

[25] C. Wan, M. C. Nnamdi, W. Shi, B. Smith, C. Purnell, and M. D. Wang, "Advancing sleep disorder diagnostics: A transformer-based EEG model for sleep stage classification and OSA prediction," *IEEE J. Biomed. Health Informat.*, vol. 29, no. 2, pp. 878–886, Feb. 2025.

[26] A. N. Olesen, P. J. Jennum, E. Mignot, and H. B. D. Sorensen, "Automatic sleep stage classification with deep residual networks in a mixed-cohort setting," *Sleep*, vol. 44, no. 1, 2021, Art. no. zsaa161.